CoverWise

AI Health Insurance Advisor

"The only advisor in the room with no commission to earn."

Agentic AI that helps Americans find their optimal ACA health insurance plan by analyzing income, medications, and doctors against live government data.

coverwise-65446133790.us-central1.run.app

Ritwik Sharan & Bhuvi Ghosh

The Problem

$1,500–$2,500

Average annual overspend from choosing the wrong plan

Kaiser Family Foundation, 2023

50–90 plans

Per ZIP on Healthcare.gov — none show true out-of-pocket cost

5+ databases

Required to make an informed decision — most people don't know they exist

THE COMPLEXITY

Navigating ACA requires cross-referencing CMS Marketplace, NPPES NPI Registry, each insurer's formulary, HRSA shortage databases, and IRS APTC tables.

Drug tier costs and prior authorization requirements are invisible to consumers.

WHAT PEOPLE DO INSTEAD

Option A: Click the lowest premium — ignores drug costs, deductibles, and out-of-pocket max exposure.

Option B: Call a broker earning **$20–$100/member/month** in carrier commissions — a direct conflict of interest.

The Market & Our Users

PRIMARY — 21.4M AMERICANS

Freelancers and independent contractors
Gig workers — Uber, DoorDash, Upwork
Self-employed business owners
People between jobs or early retirees

SECONDARY — 160M AMERICANS

Employer-covered workers facing open enrollment with no personalized guidance on plan selection

$1.2T
annual premiums

60 days
enrollment window

PRIMARY PERSONA

Maria, 34 — Chicago Freelance Designer

Earns $52,000/year · Ozempic + Metformin · Endocrinologist quarterly

- Every November she spends **4–6 hours** on healthcare.gov unsure whether a Bronze plan at **$7,400 deductible** beats Silver at **$3,000.**
- She still cannot confirm if **Ozempic is covered** or her endocrinologist is **in-network.**
- She qualifies for a **CSR Silver subsidy** that cuts her deductible in half — she has never been told this.
- **Gig workers, self-employed, between jobs, early retirees** — all face the same problem with no HR team to help.

She has made the wrong call for three consecutive years.

Why CoverWise

DRUG-LEVEL ACCURACY

Cross-references exact medications via RxNorm + CMS formulary. Surfaces tier costs and prior authorization requirements before enrollment.

DOCTOR VERIFICATION

Verifies each doctor's active status and network inclusion via NPPES NPI Registry. No more surprise out-of-network bills.

SUBSIDY CALCULATION

Calculates exact APTC and CSR eligibility using live HHS FPL tables and household size. Most users don't know what they qualify for.

EV-RANKED RECOMMENDATIONS

Plans ranked by Expected Value weighted to the user's actual utilization pattern — not just premium cost. Lowest EV wins.

THE STRUCTURAL DIFFERENCE

No commissions. No conflict of interest. Every figure from live government APIs before any AI reasoning. A year-round advisor — not just November enrollment.

Business Model — Freemium

FREE

$0

forever

- ✓ 3 plans shown
- ✓ 1 drug check
- ✓ 1 doctor check
- ✓ EV ranking + AI recommendation
- ✓ Chat advisor
- ✓ Specialist finder
- ✓ Procedure estimator
- ✗ HSA 5-year wealth forecast

Great for first-time users

PREMIUM

BEST VALUE

$19/mo

billed monthly

- ✓ **10 plans** shown
- ✓ **Unlimited** drug checks
- ✓ **Unlimited** doctor checks
- ✓ Deeper AI recommendation
- ✓ Chat advisor with full context
- ✓ Specialist finder
- ✓ Procedure estimator
- ✓ **HSA 5-year wealth forecast**

For freelancers managing their full financial health

System Architecture — Multi-Wave Pipeline

Wave 0
ZIP → FIPS → State
Location routing

→

Wave 1
asyncio.gather
Subsidy + Plan search

→

Wave 2
asyncio.gather
Drug · Doctor · Risk

Phase 1.5 — Python EV Ranking
Instant · Deterministic · Zero LLM cost

Phase 2 — Gemini 2.5 Flash
Single synthesis call · 4-pillar recommendation

FPL Range	Routing Decision
< 138%	Medicaid Agent — free coverage path
138–400%	ACA Marketplace with APTC / CSR subsidies
> 400%	ACA Marketplace — full premium
NY / CA / MA	State Exchange Redirect

Engineering Concepts

Every dollar figure is real data — all six APIs are called before any LLM reasoning occurs.

CMS Marketplace	NPPES NPI	RxNorm (NIH)	openFDA	HHS FPL	CMS Eligibility
Plans, formulary, network	Doctor identity	Drug → RxCUI	Generic alternatives	Subsidy thresholds	APTC / CSR

1. Conversational Intake Agent

Separate ADK agent collects ZIP, age, income, medications, and doctors via natural language before the pipeline fires. `/api/intake/start` · `/api/intake/message`

2. Multi-Agent Orchestration

Three parallel waves — **~10–15s** latency despite 6+ APIs.
`adk_orchestrator.py`

3. Agentic Memory

mem0 + ChromaDB stores profile post-analysis for year-round advising.
`mem0_client.py`

4. Human-in-the-Loop Gate

User confirms extracted profile before pipeline fires. Analysis only starts on approval.

5. Google ADK — Year-Round Advisor

ADK tool-calling selects the right API at runtime — plan search, drug lookup, subsidy check, specialist finder, enrollment dates.

EV RANKING FORMULA

```
EV = (w_h × Healthy) + (w_c × Clinical) + (w_w × Worst)
```

Pure Python — zero LLM cost, instant, deterministic.

CSR override: FPL 138–250% → Silver always rank 1

Technology Stack

AI & ORCHESTRATION

Google ADK — agent framework and runtime tool-calling

Gemini 2.5 Flash — single synthesis LLM call per analysis

asyncio — parallel wave orchestration

DATA & MEMORY

mem0 — agentic memory, profile caching & cross-session retrieval

ChromaDB — vector store backing mem0 for semantic lookup

RAG Formulary Index — insurer MRF JSON seeded into ChromaDB at startup; drug tier lookups served from index before falling back to CMS API

CMS · NPPES · RxNorm · openFDA · HHS — live government APIs

BACKEND & INFRA

FastAPI — REST API layer

Google Cloud Run — serverless (4 GiB / 2 vCPU)

Python EV engine — deterministic ranking, zero LLM cost

Key design decision: EV ranking is pure Python arithmetic — the most compute-intensive step has zero LLM cost and sub-millisecond latency.

LLM usage: One call per analysis (synthesis only). All data collection is deterministic API calls — no LLM reasoning until the full picture is assembled.

Token Economics & Unit Economics

INPUT	OUTPUT	THINKING
$0.075	**$0.30**	**$3.50**
per 1M	per 1M	disable it

Component	Tokens	Cost
System instruction	~975	$0.000073
User data doc	~3,025	$0.000227
Output — recommendation	~1,500	$0.000450
Total per /api/analyze	**~5,500**	**$0.000750**

Chat: **$0.00040**

Q&A: **$0.000113**

Premium user — 1 analysis + 8 chats + 4 Q&As/month

Item	Cost
Total LLM	$0.0044
Cloud Run	$0.040
mem0 + ChromaDB	$0.005
Government APIs	$0.000
Total cost to serve	**$0.050**
Revenue	**$19.00**
Gross margin	**99.7%**

10x power user (25 analyses, 100 chats) costs under **$0.30** in tokens/month.

Risks & Growth Path

WHERE THE MODEL BREAKS

Thinking tokens: If Gemini fires a thinking budget, output cost jumps from $0.30/M to $3.50/M. Fix: `thinking_budget: 0`

Seasonality: Open enrollment is Nov 15–Jan 15 (60 days). LTV = $19–$38. At $30–$50 CAC via paid ads, the model is break-even. Year-round advisor is the retention play — unproven.

Conversion math: 3% free-to-premium, 60% churn → LTV ~$32. Only viable if CAC < $32 — organic/SEO only.

Untapped revenue: Broker commissions = $20–$100/member/month. Licensed referral fees could **5–10x LTV**.

PATH TO $100K ARR

440
premium subscribers

15K
free users at 3% conversion

60–70%
sign-ups in Nov–Jan

Metric	Value
Fixed cost floor	~$15/month
Breakeven	1 premium user
LTV (60% churn)	~$32
Max viable CAC	$32 (SEO only)

Primary acquisition: SEO on "ACA plan comparison", "health insurance calculator", "Obamacare subsidy estimator"

Competitive Landscape & Risk Register

Capability	CoverWise	healthcare.gov	eHealth.com	Local Broker
Personalized drug analysis	Yes	No	No	Sometimes
Doctor NPI verification	Yes	No	No	Rarely
Subsidy calculation	Yes	Yes	Yes	Yes
No commission conflict	Yes	Yes	No	No
Year-round AI advisor	Yes	No	No	No
Price	Free / $19/mo	Free	Free (lead-gen)	Free (commission)

Risk	Mitigation
CMS API availability	TTL cache absorbs ~75% of calls; graceful degradation returns data-only results
LLM cost spike at scale	Gemini 2.5 Flash pricing; hard per-user-tier rate limits in API layer
Liability — not a licensed broker	Disclaimer on every output: "Informational only. Verify on healthcare.gov."
Open enrollment seasonality	Year-round advisor + mid-year SEP guidance smooth monthly active users

Summary

WHAT WAS BUILT

6-API parallel agent pipeline with ~10–15s end-to-end latency

Deterministic Python EV ranking — zero LLM cost, instant, auditable

Persistent user memory via mem0 + ChromaDB for year-round advising

Human-in-the-loop confirmation gate before compute is spent

Google ADK Q&A advisor with runtime API routing

BUSINESS FUNDAMENTALS

Gross margin: 99.7% per premium user

1,000 free users: $1.78/month in LLM costs

Breakeven: 1 premium subscriber

Path to $100K ARR: 440 subscribers via organic SEO

THE THESIS

Americans overspend $1,500–$2,500/year on health insurance not because the information doesn't exist — but because no one has assembled it into a conflict-free, personalized recommendation. CoverWise does exactly that, at **$0.00075 per user.**

Live Deployment

`coverwise-65446133790.us-central1.run.app`

Branch: adk-insurance-advisor-release